KAIST

HUMAN LAB
HUMAN-MACHINE INTERACTION

인간-기계 상호작용 연구실

Human Machine iNteraction

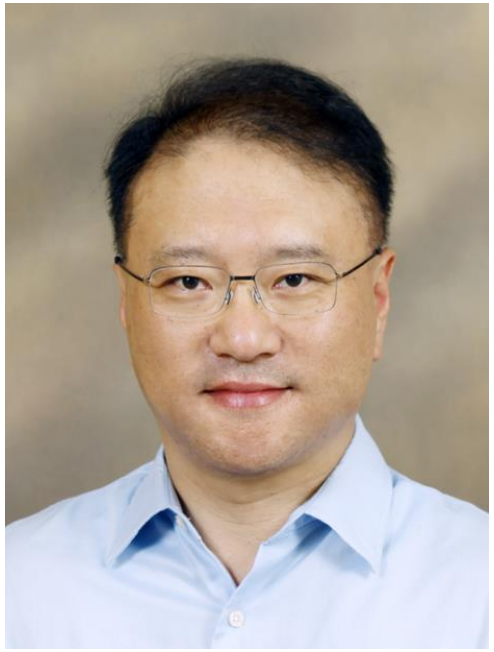


As of August 10th, 2021

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yhpark@kaist.ac.kr
Human-Machine iNteraction LAB
Dept. Mechanical Eng., KAIST

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교수와 랩 소개



Prof. Yong-Hwa Park

(1987~1999) BS, MS, Ph.D at Mechanical Eng., KAIST
Structural Vibration and Optimization

(2000~2003) Post Doc. of Univ. of Colorado at Boulder
Adv. Structural Dynamics, Micro-opto-mech system

(2003~2016) Research Master of Samsung Electronics
Sensor Device and Systems (3D Lidar, health sensors.)
Published 70+ US Patents

KAIST 기계공학과, 인공지능연구소 부교수
KAIST 기계공학과 산학협력위원장
SPIE Photonics West 학회좌장
대한기계학회, 소음진동공학회 이사

Tennis Lover!

1996, 2000 KAIST Championship!

Research Interests

Human-Machine Interactions:
3D *Vision, Acoustic Recognition, Health Sensors*
Vibration, Structural Health Monitoring
Machine Learning

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Perceptive Human-Machine iNteraction with Smart Sensors



Human

우리 친해지고 싶어요~



Machine

"기계가 인간처럼 **시각 청각 촉각**을 통해 세상을 **인식할 수 있도록 하는 센서와 인식**에 대해 연구합니다."

2021년 가을 기준 석사과정 7명, 박사과정 13명, 연구원 1명

주인공 토니스타크를 도와주는 **인공지능 집사 Jarvis**



[스타크]
"자비스, 심장 체크해봐. 내 뇌가 이상한거야?"

[자비스]
"심장이랑 뇌에는 전혀 문제가 없어요."

[스타크]
"그럼, 독극물에 중독됐어?"

[자비스]
"제가 보기엔 심각한 불안증세예요."

My Motivation!

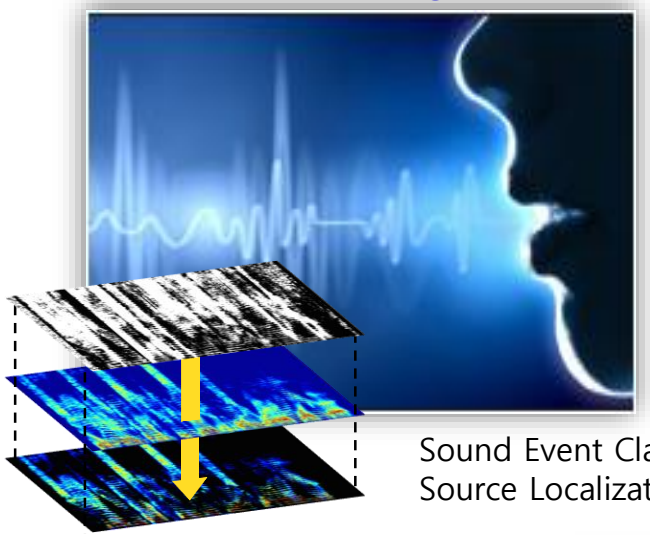
Research Portfolio

3차원 공간 인식
3D scene recognition



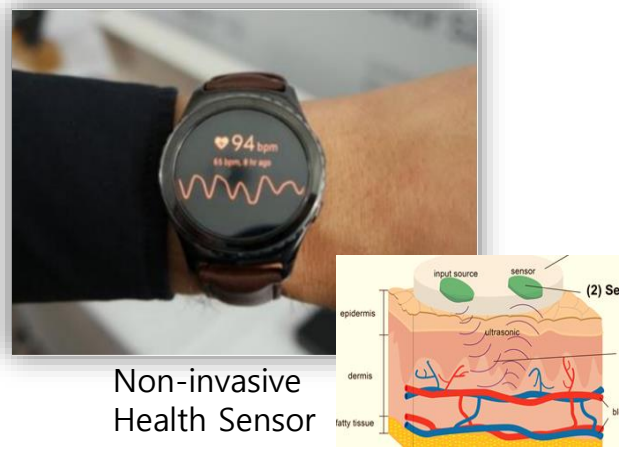
3D 공간 Map

음향 인식
Acoustic Recognition



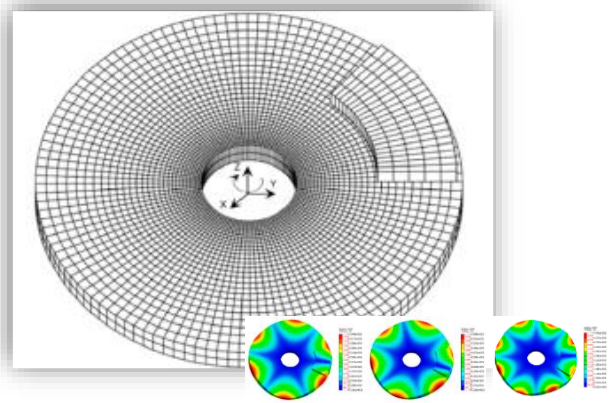
Sound Event Classification
Source Localization

생체 인식
Biometric Recognition



Non-invasive
Health Sensor

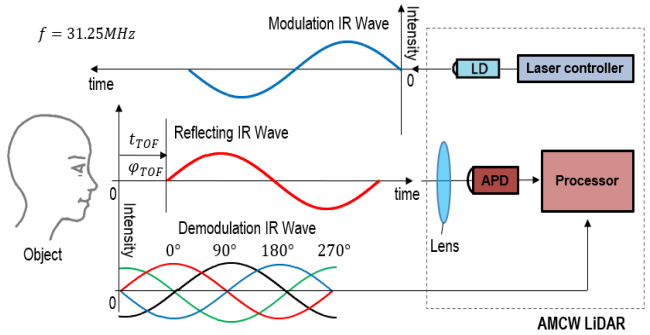
진동 해석.인식
Vibration Analysis & Recognition

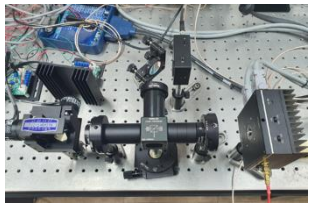


Hi-fidelity Vibration Analysis
Structural Health Monitoring

3차원 공간인식 3D Scene Recognition

초소형 고정밀 3D Sensor (Lidar)



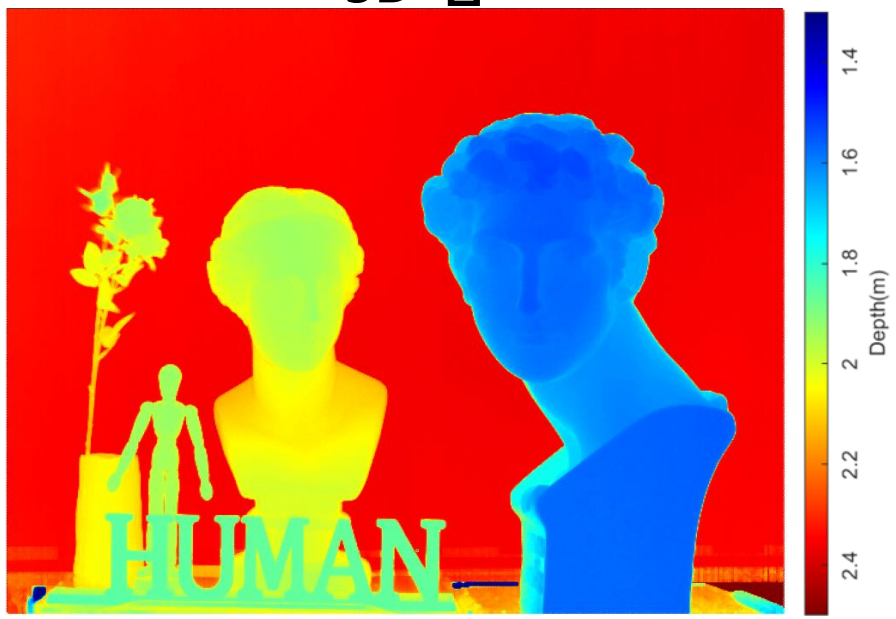


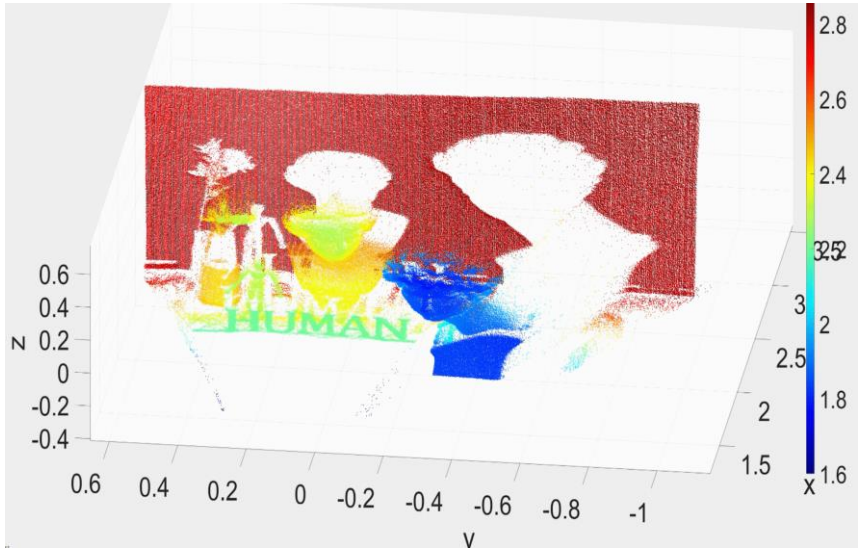
Continuous Wave
Parallel Demodulation
3D Scanning Lidar

Scene

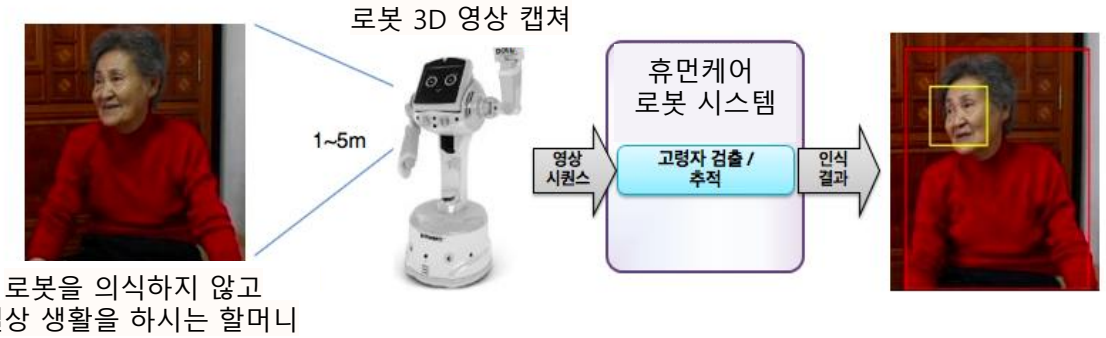


3D 맵





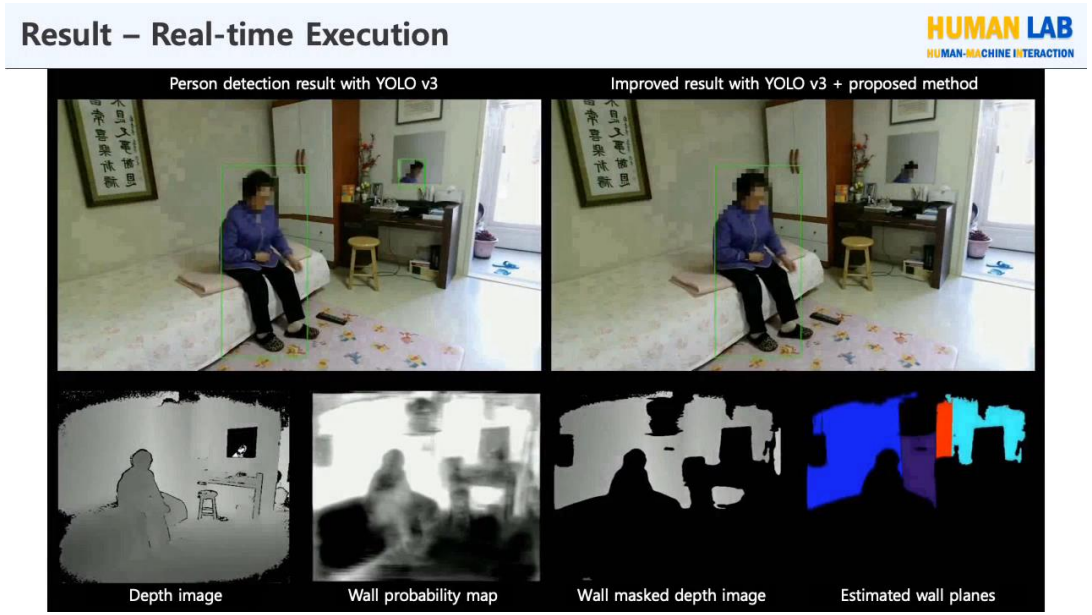
휴머노이드 로봇



로봇을 의식하지 않고
일상 생활을 하시는 할머니

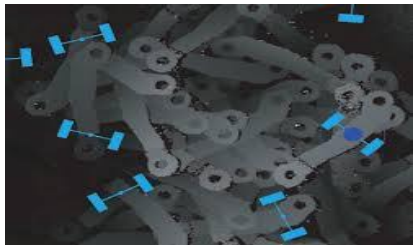
자율로봇 3차원 물체인식





로봇 제조 공정 (bin picking)



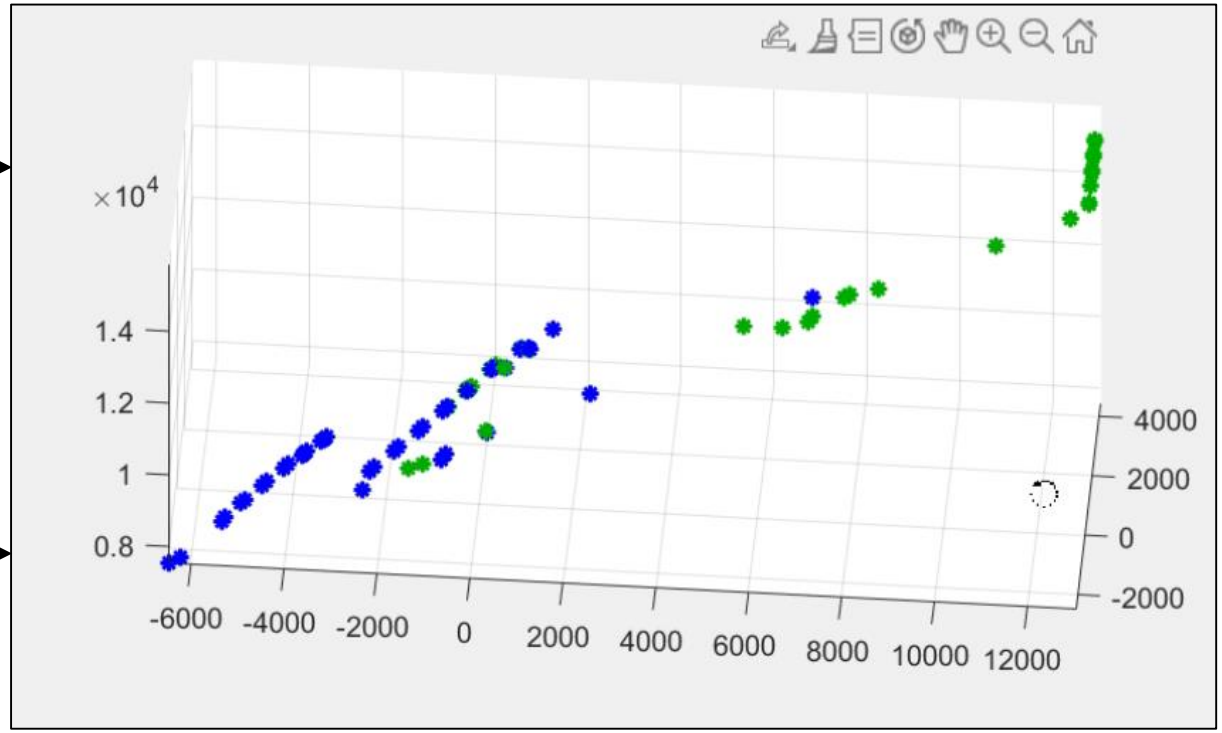




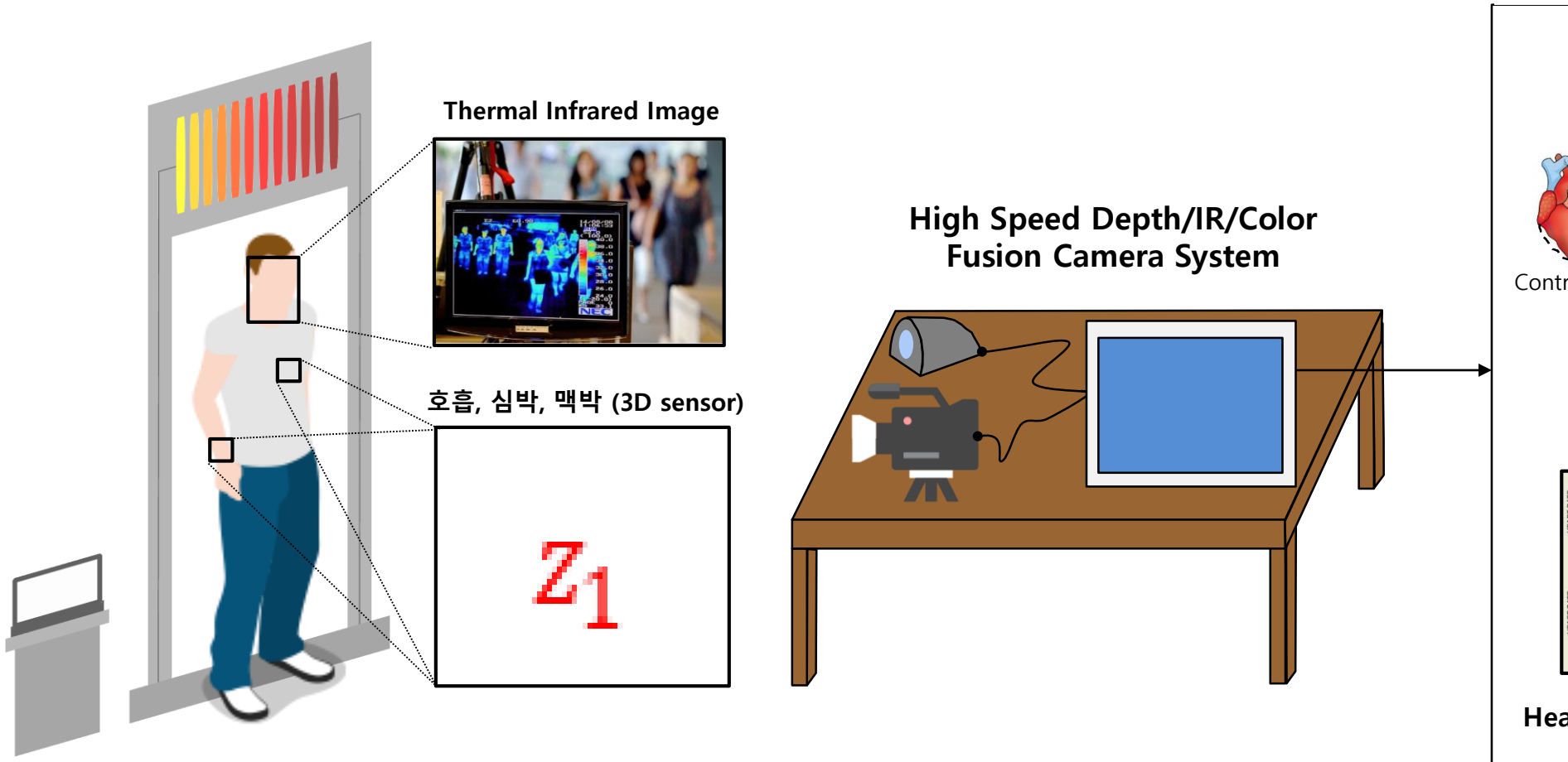




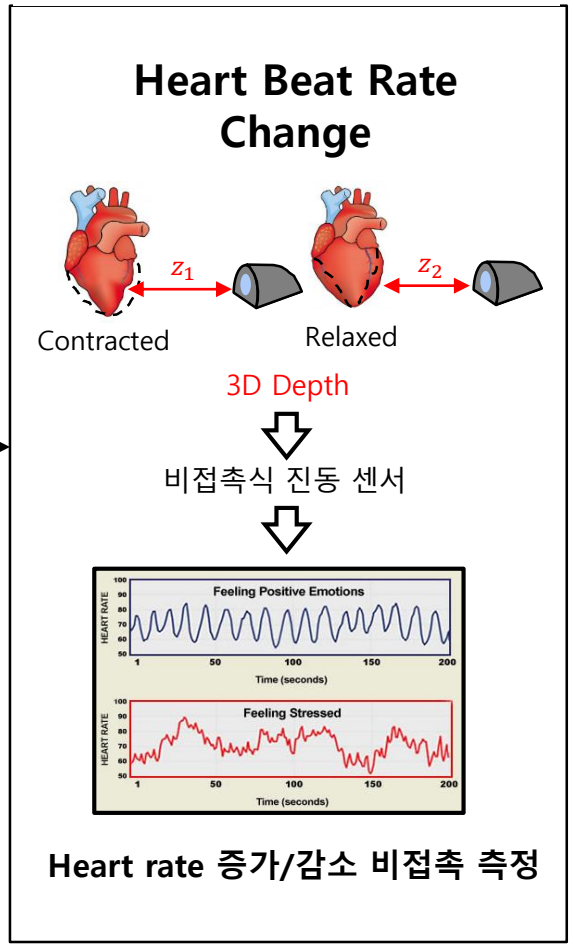
스테레오 기반 대규모 제조 공정의 정밀 3D 좌표 인식



Biometric Recognition with Remote 3D Sensing (Untact)



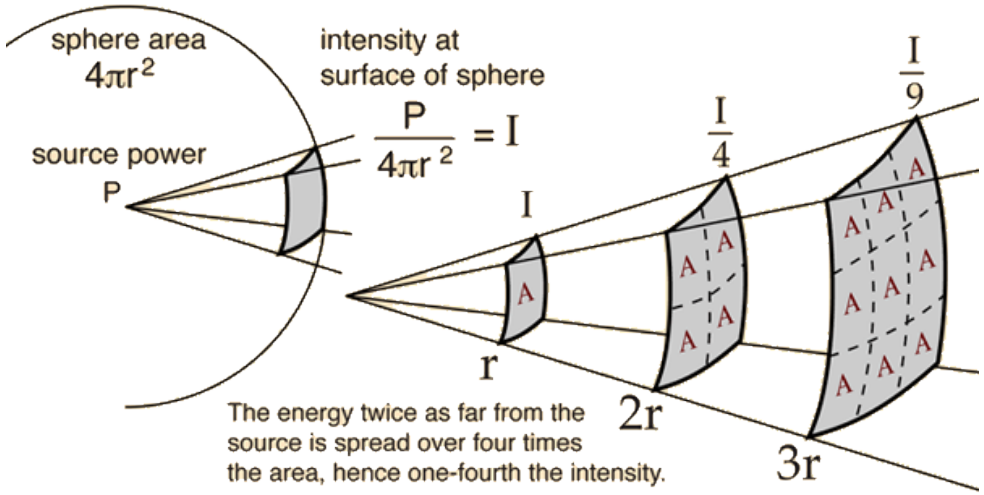
원거리 생체 신호 감지
(호흡, 맥박, 심박)

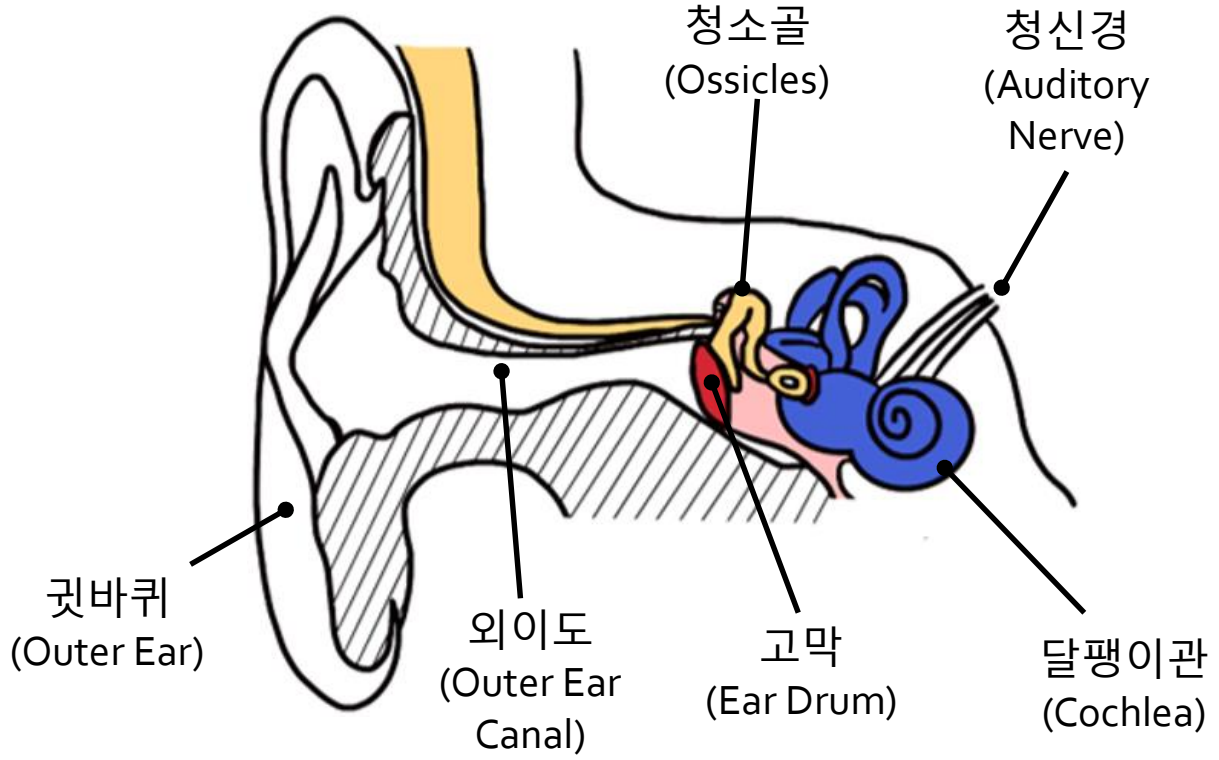


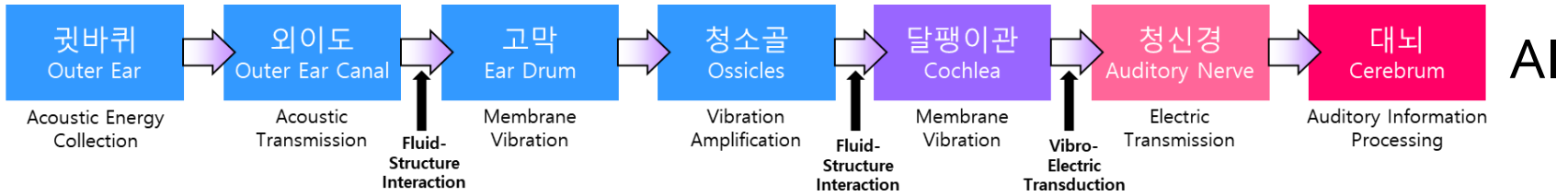


A typical cocktail party scene (image from Daniel Handerman: **High Society Cocktail Party – End of Prohibition 1933**)
from Cherry EC, 1953. Some experiments on the recognition of speech, with one and with two ears. J Acoust Soc Am, 25(5):975-979.

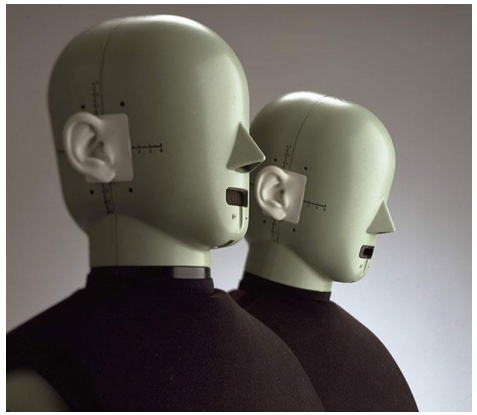
공간 음향 + 인간의 청각계 + 대뇌 인지 (AI)

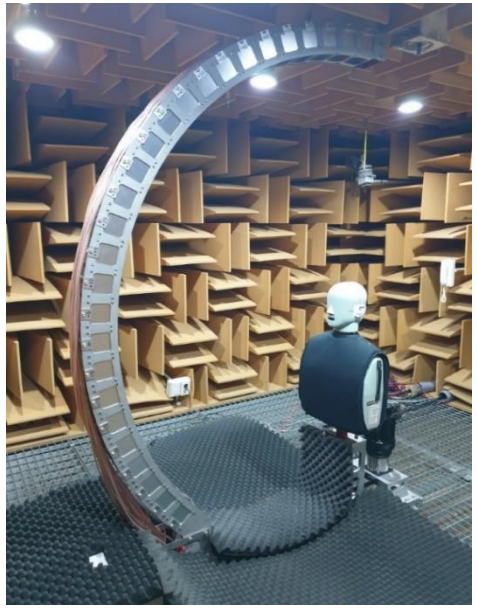




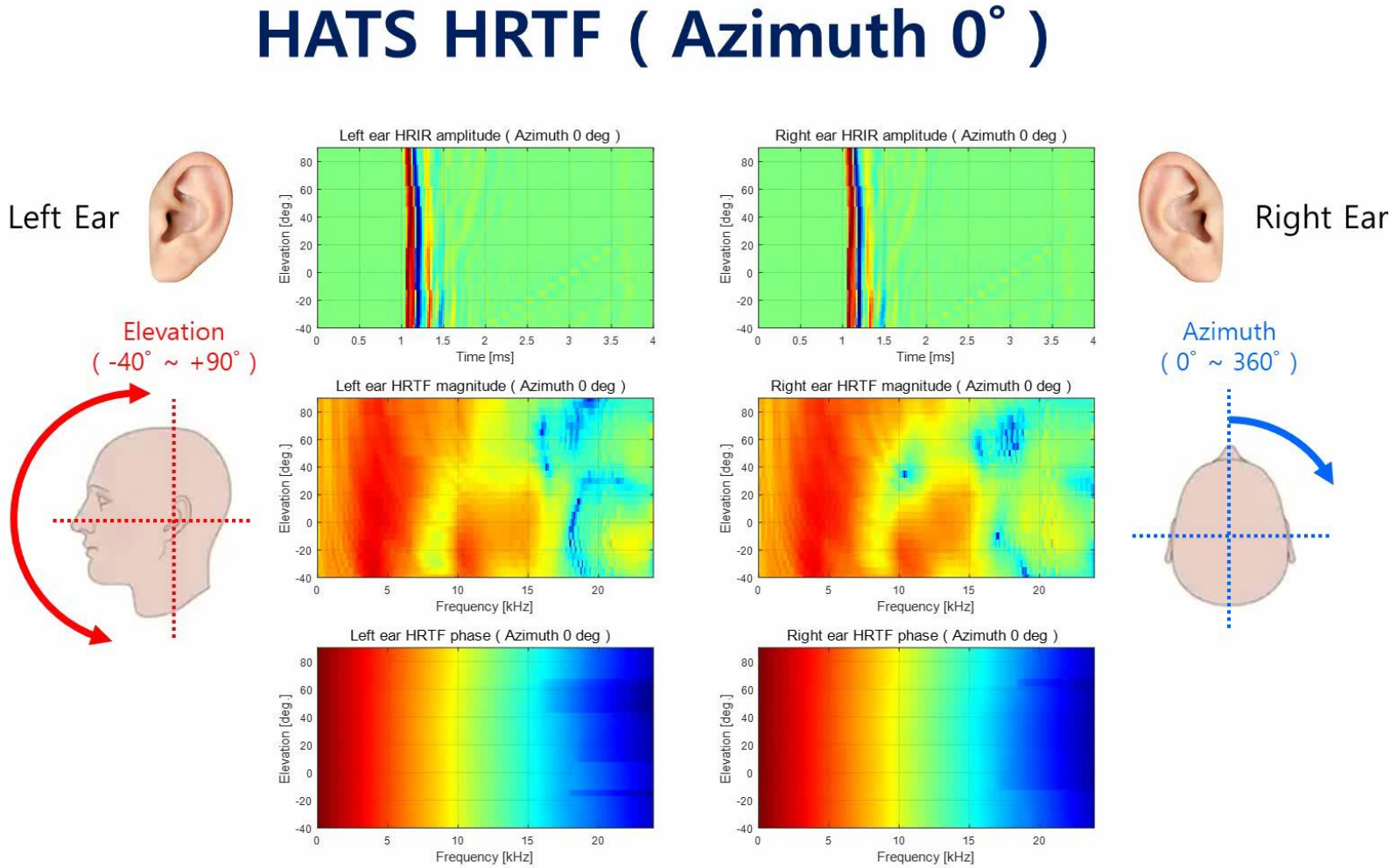


머리전달 함수 - Head Related Transfer Function

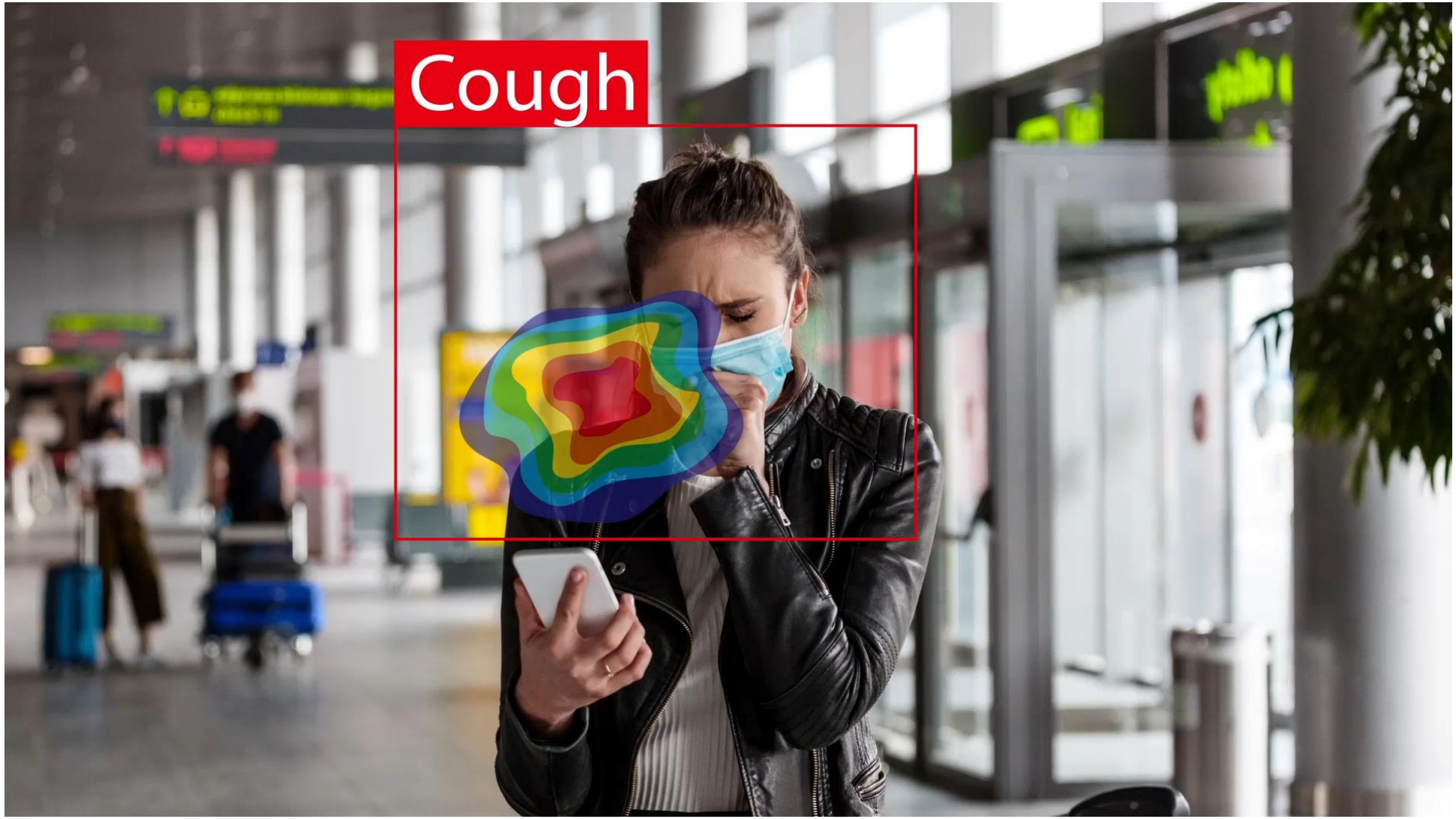




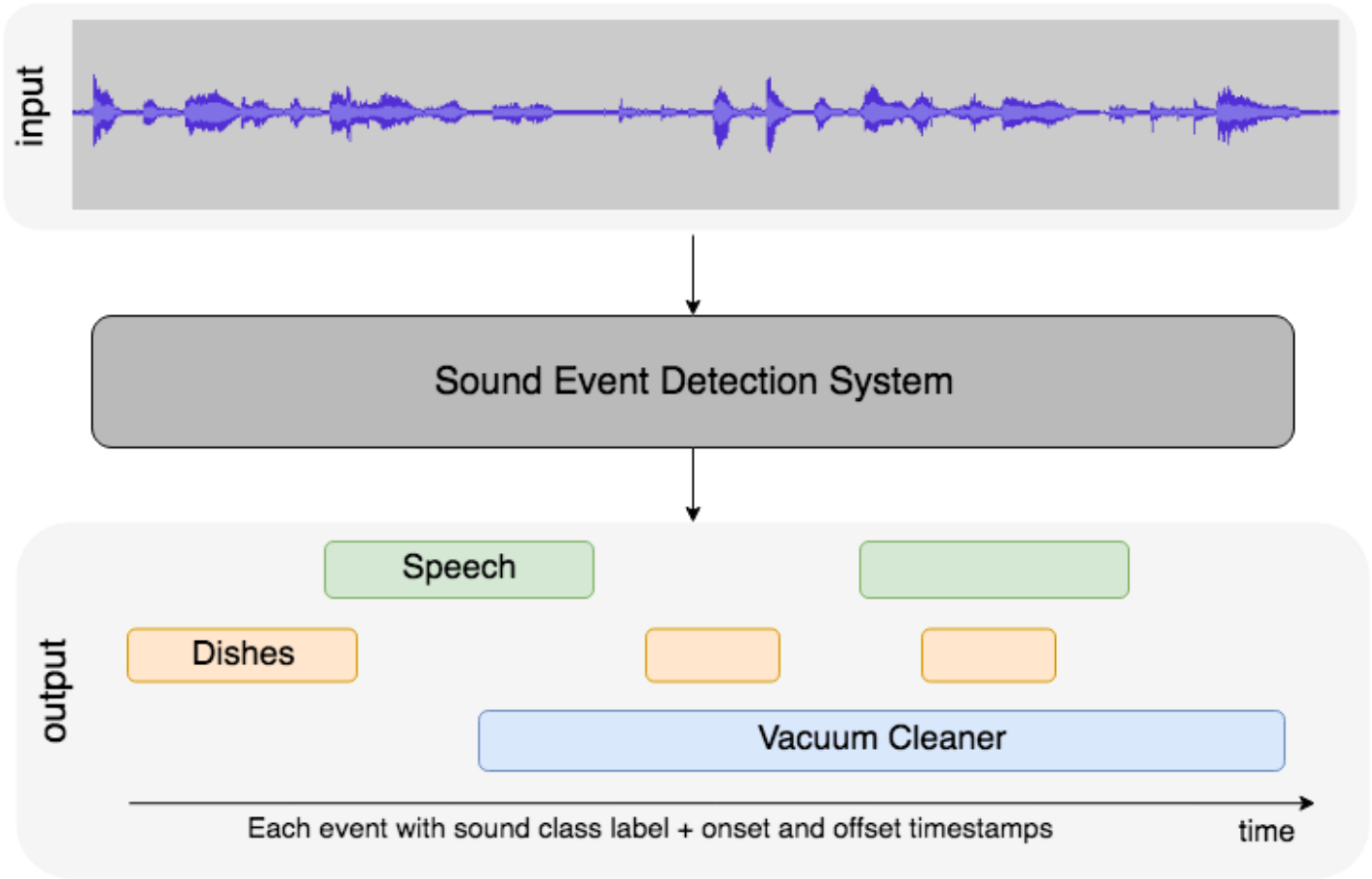
NOVIC+, Human Lab 무향실



KAIST 기침인식 카메라 (Cough Detection Camera)

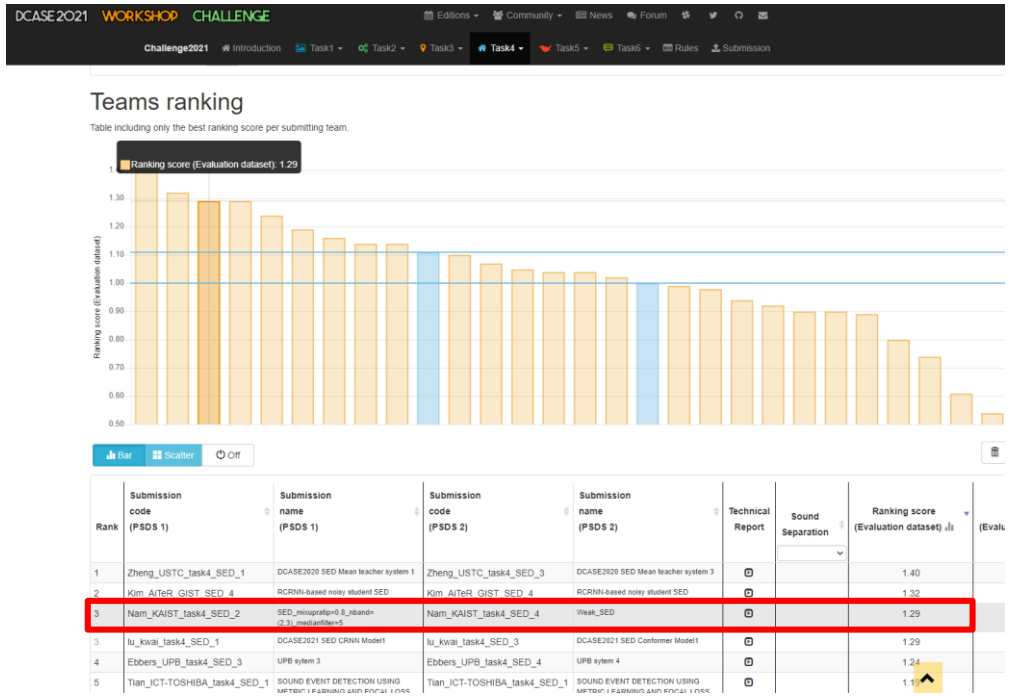


음향 이벤트 인식(Sound Event Detection)



IEEE DCASE 2021 Challenge Task4

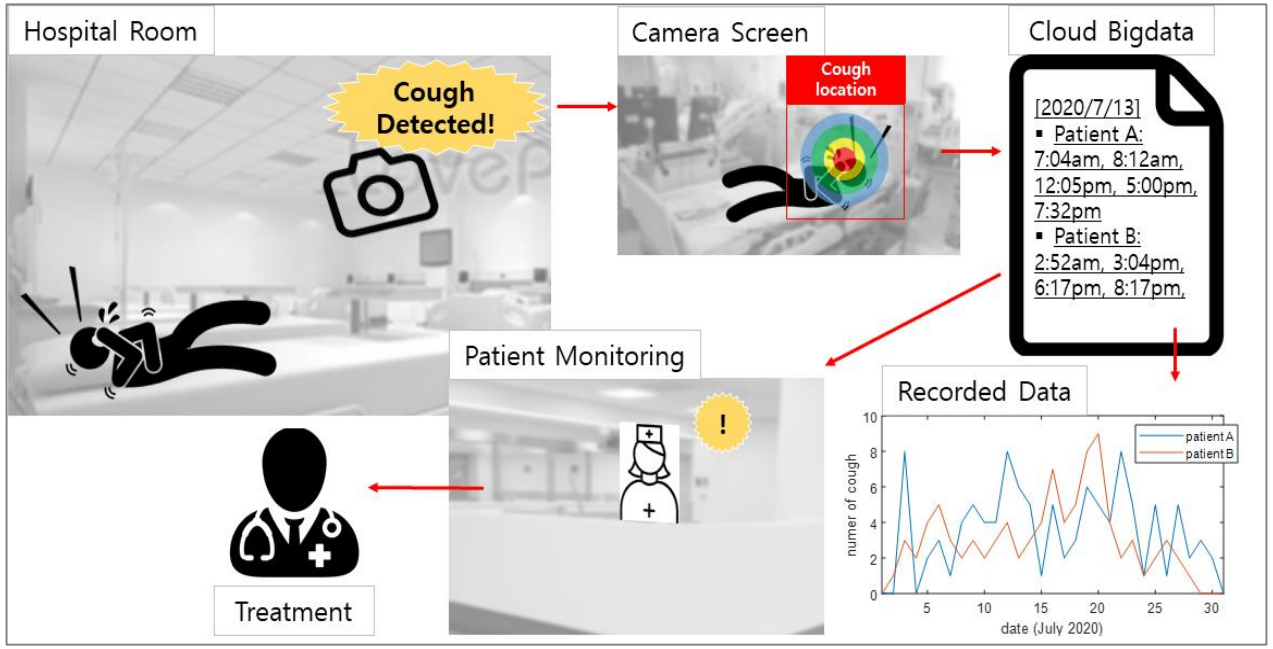
- AI를 활용하여 특정 오디오파일 내에 몇초부터 몇초 사이에 어떤 음향이벤트가 존재하는지를 인식
- 총 24개의 팀 중 3위
- Arxiv 논문 게재 및 Github 코드 공개



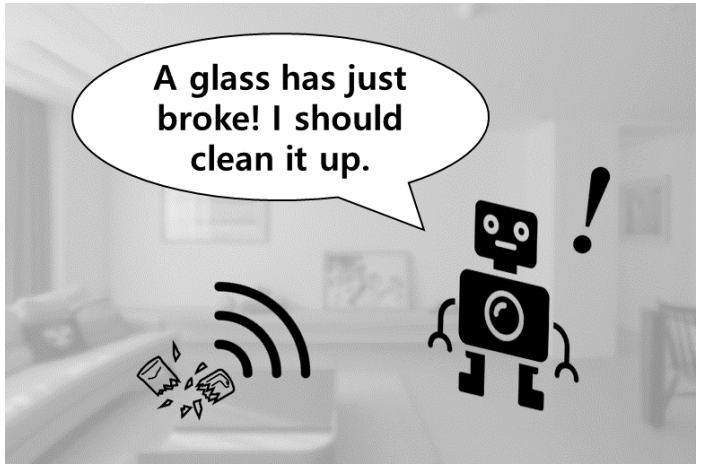
병원 환자 돌봄 (학교, 식당 ....)



- 기침소리를 인식하고 환자와 빈도를 실시간 모니터링



휴먼 케어 로봇 - 환경음향인식

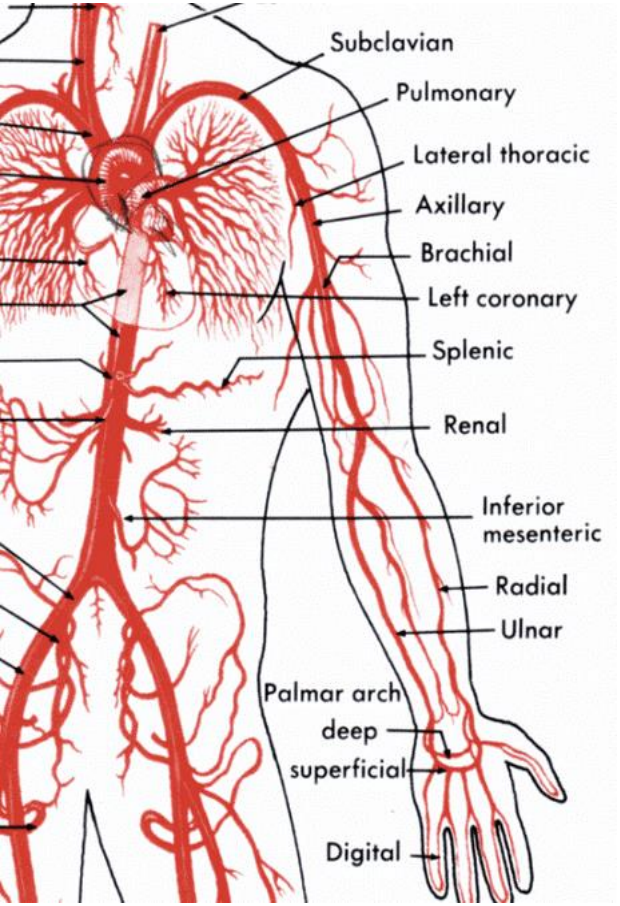


기침
코골이
낙상
깨지는 소리
화재경보
침입
초인종
핸드폰 알람

음향 기반 고장 진단

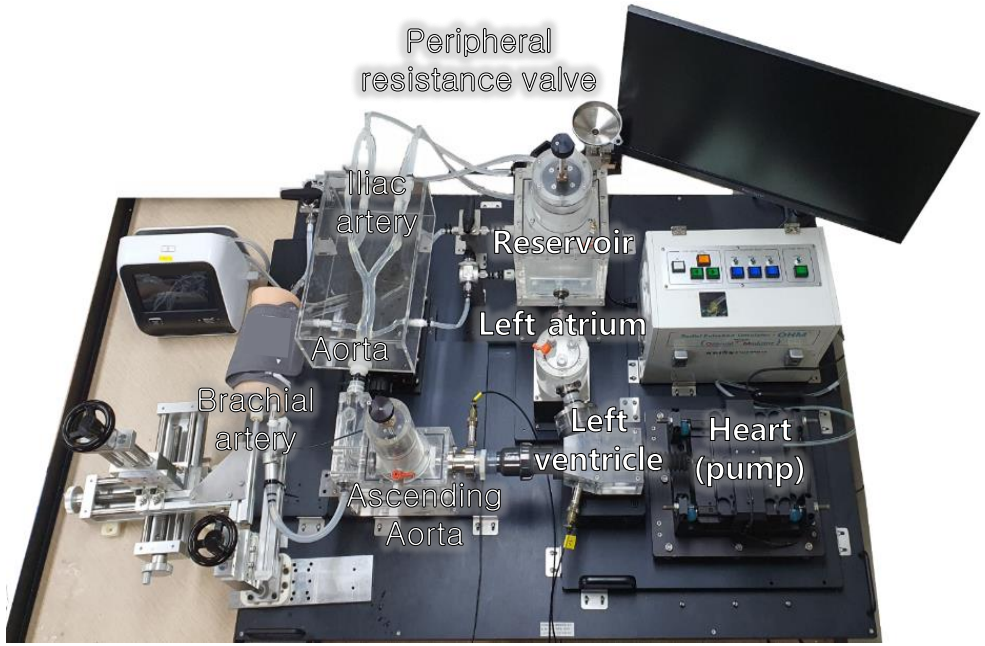


Virtual Patient - 심혈관계 질환 가상환자

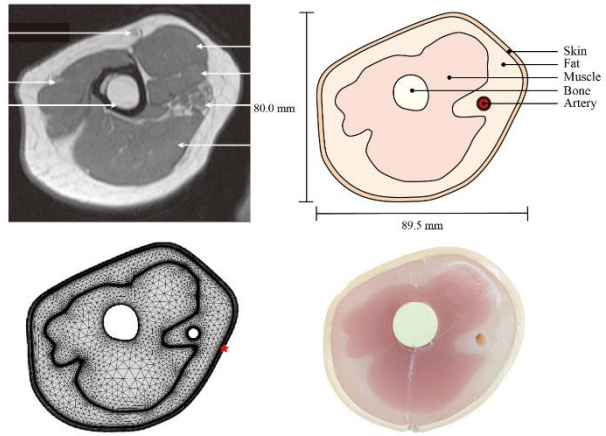


Virtual Patient for Cardiovascular Studies

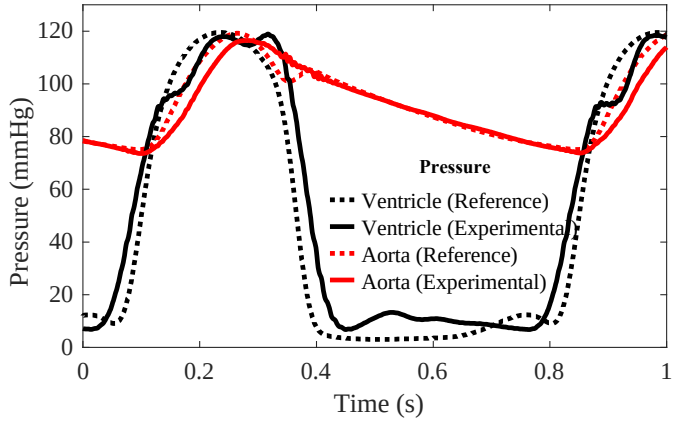
- Artificial cardiovascular system mimicking human body for study of cardiovascular disease (virtual patient)
- In depth study on the correlation between CVD and human organ and tissues in virtual setup



Artificial Cardiovascular System in Lab

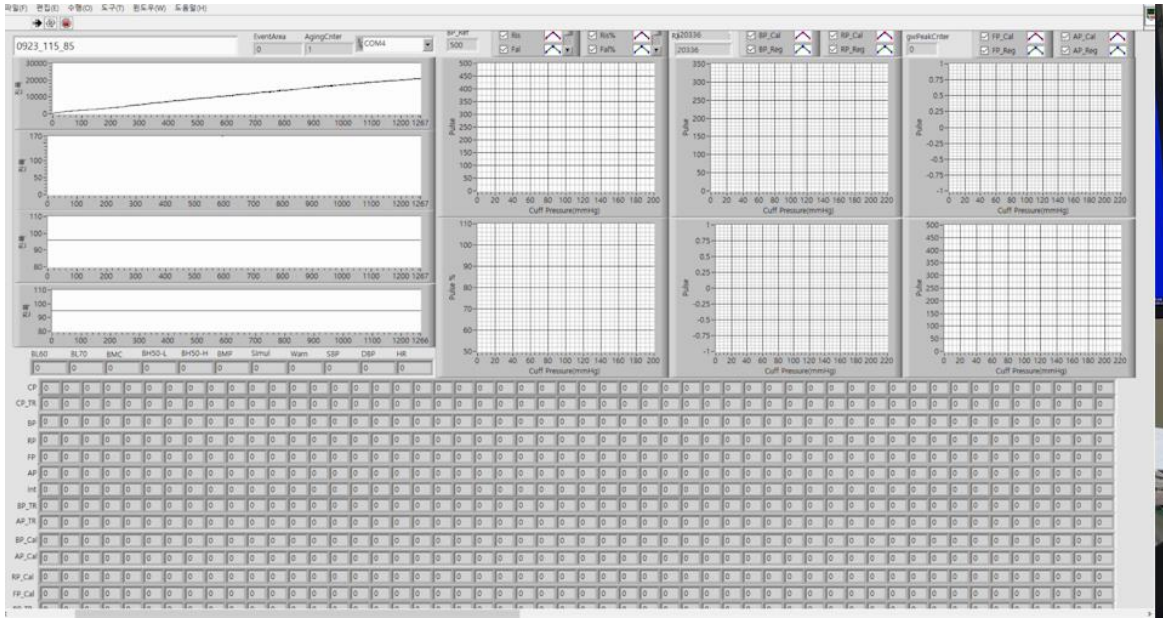


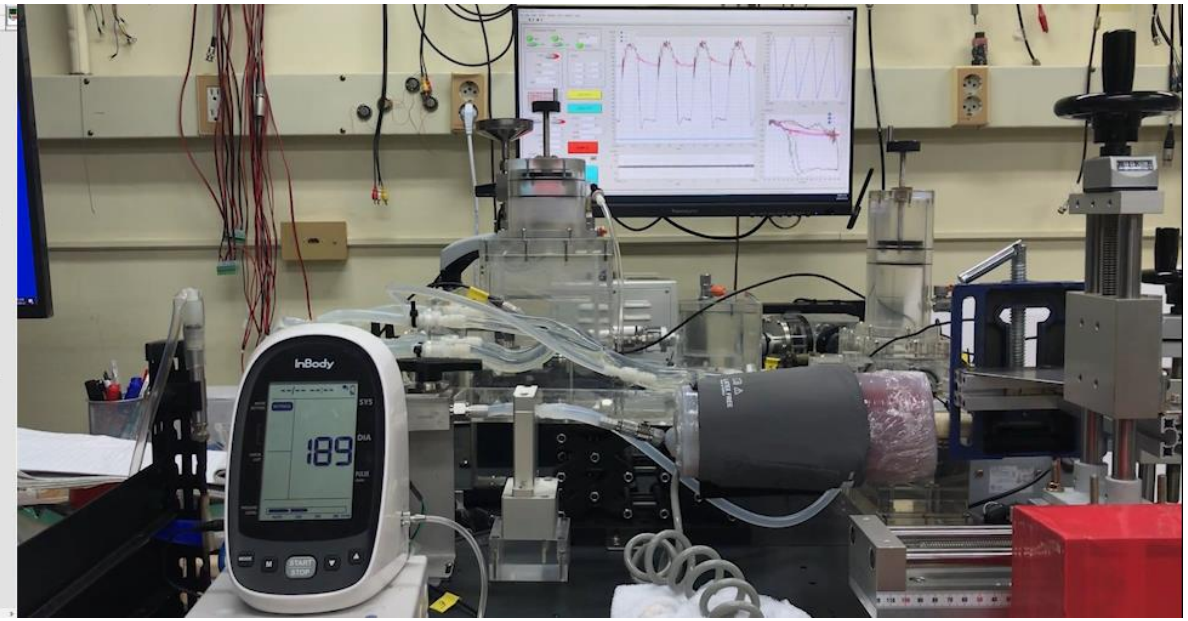
Reconstruction of blood pressure transmission mechanism in upper arm
: MRI image based numerical model and phantom for blood pressure analysis

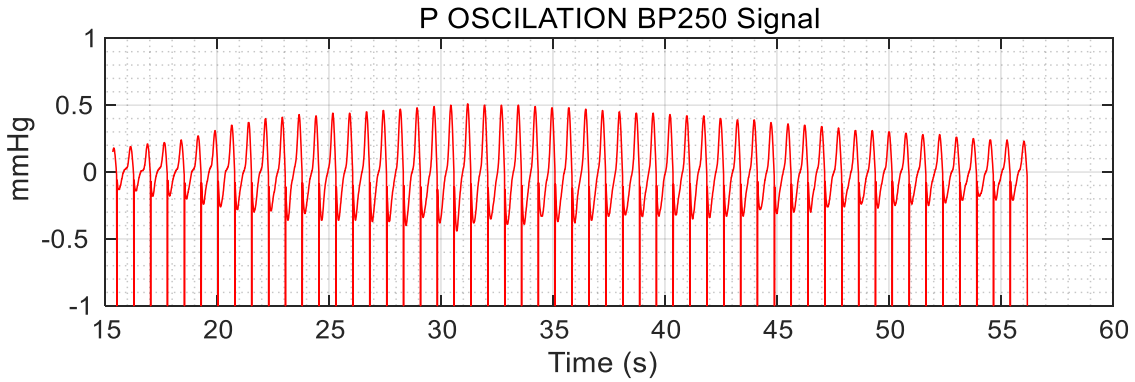


Reproduction of human blood pressure
: Ventricular and aorta pressure compared with Wigger's diagram (reference)

가상환자 시스템을 이용한 상용 혈압계 측정 영상



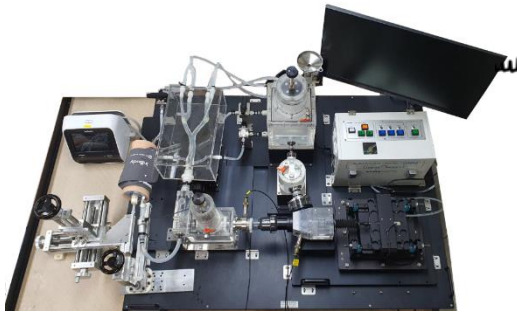


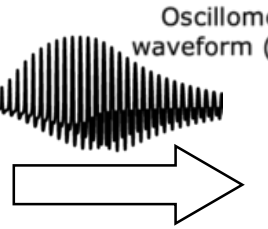


- 인체와 유사한 크기의 오실로메트릭 (혈압계) 신호 관측
- 인체와 유사한 맥박 및 맥파 관측 가능
- 혈압 측정치 오차 개선 중

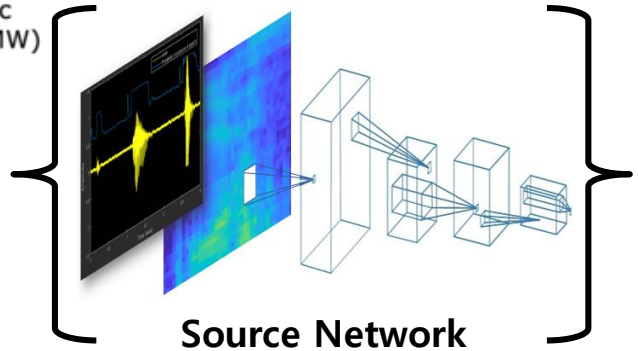
혈압계를 이용한 심혈관계 상태 진단 (심장, 대동맥, 혈관벽 등)

Virtual Patient
(Source Domain)





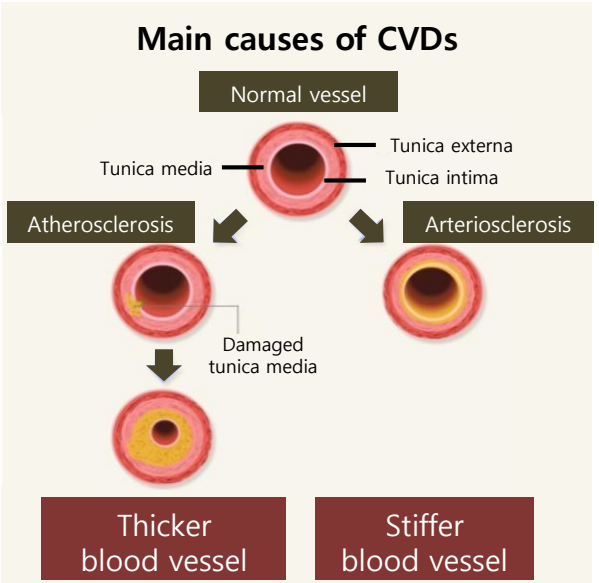
다양한 심혈관계
상태 모사 후 혈압
신호 측정



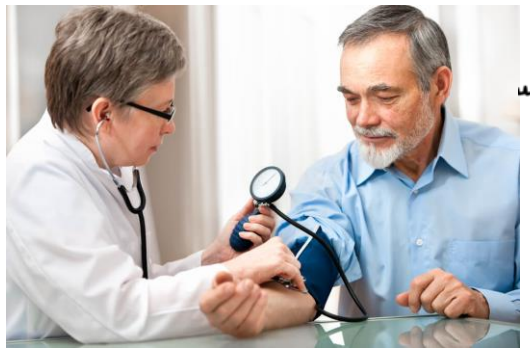
Source Network

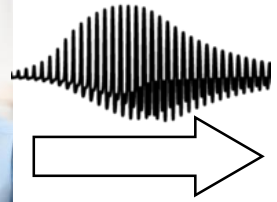
심혈관계
상태재현

혈관나이
동맥경화
심박출량
대동맥류

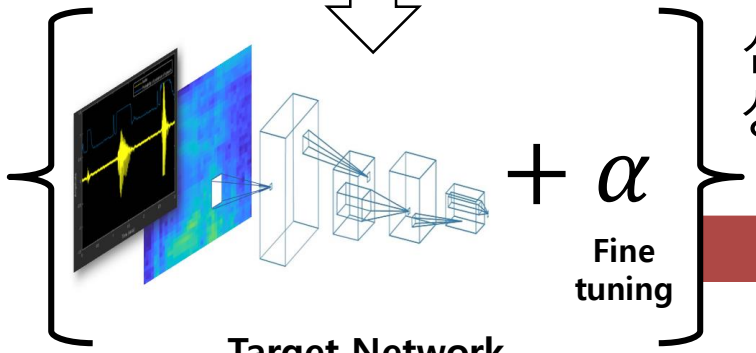


Human
(Target Domain)





"Transfer Learning"



Target Network

심혈관계
상태진단

심혈관계 상태 모니터링이 가능한 혈압계



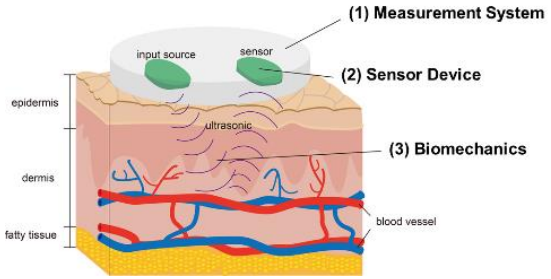
비침습 혈당센서 – Non-invasive Glucose Sensor by Human Body Vibration

Non-invasive glucose measurement sensing architecture (patented)

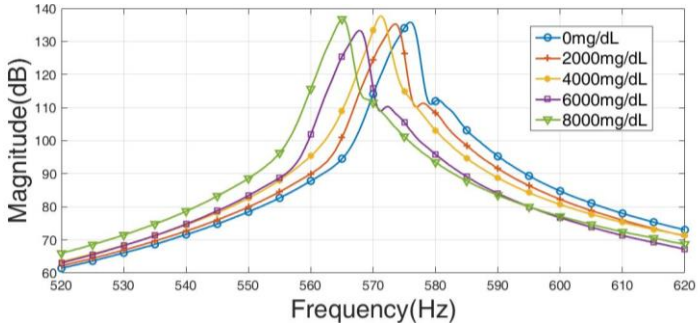




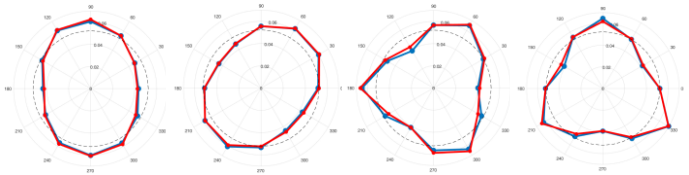




Frequency response function(2nd mode) and Glucose Concentration



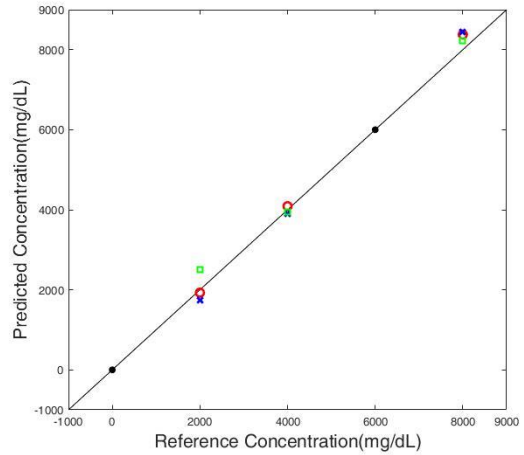
Mode shape analysis of sensor



Mode	1st	1st(2)	2nd(1)	2nd(2)
MAC	0.99	0.99	0.94	0.96

US/KP Patent

Glucose concentration extraction result



생체 인증 - Biometric Authentication using Human Body Vibration

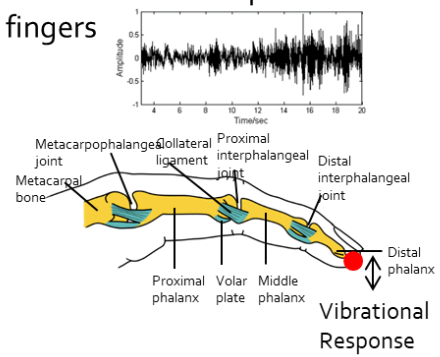
인체 진동을 이용한 모바일 기기 생체인증 시나리오



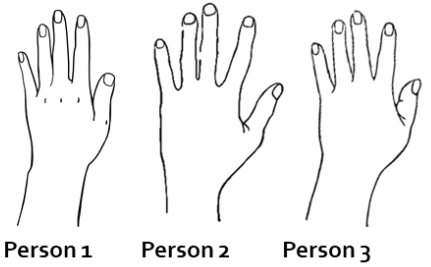
① Contact the device with fingers

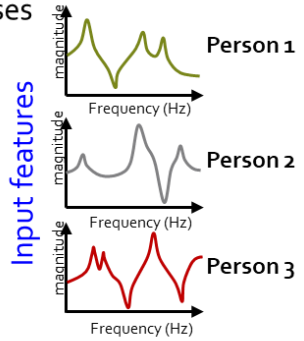


② Vibration transducer measures the vibrational response of fingers

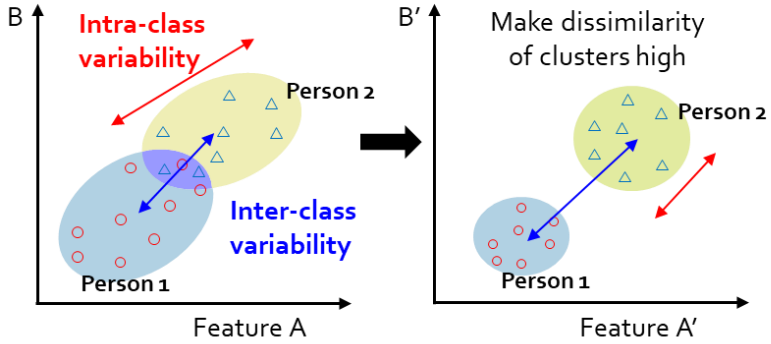


③ Calculate the frequency responses of fingers or different person

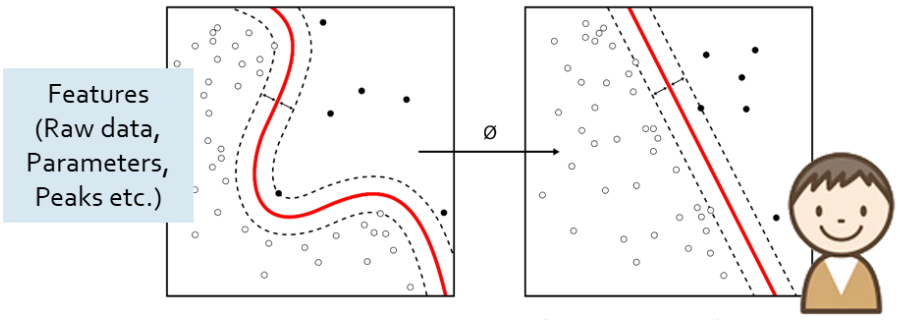




④ Optimize features which maximize the inter-class variability and minimize the intra-class variability

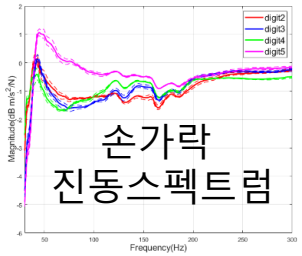


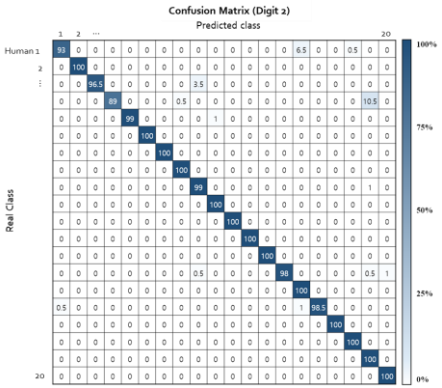
⑤ Apply the classification algorithms (SVM)



생체인증 실험 및 결과



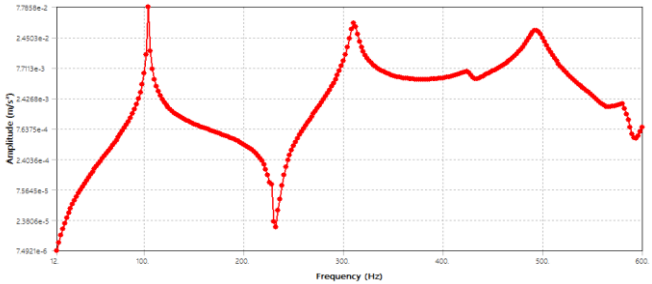




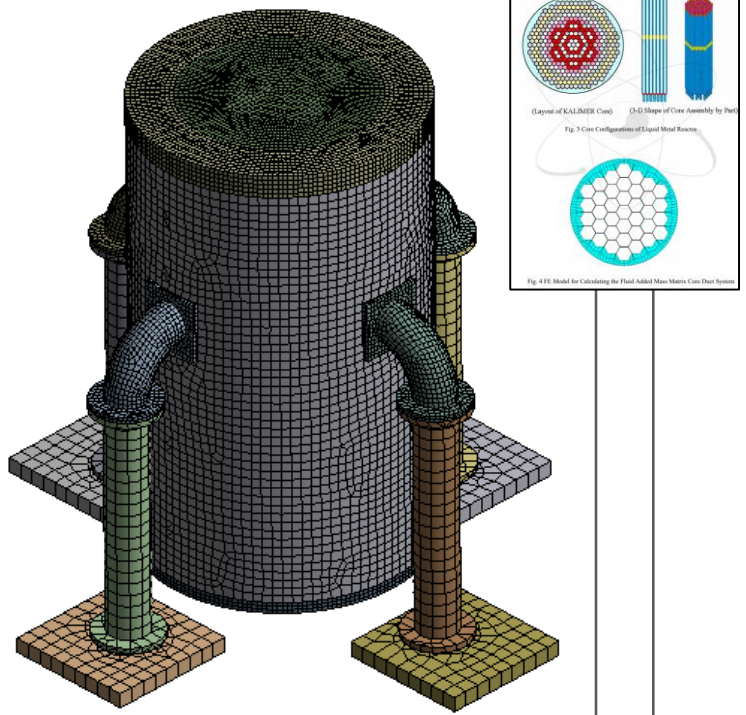
개인구분 정확도 99%

High Fidelity Digital Twin

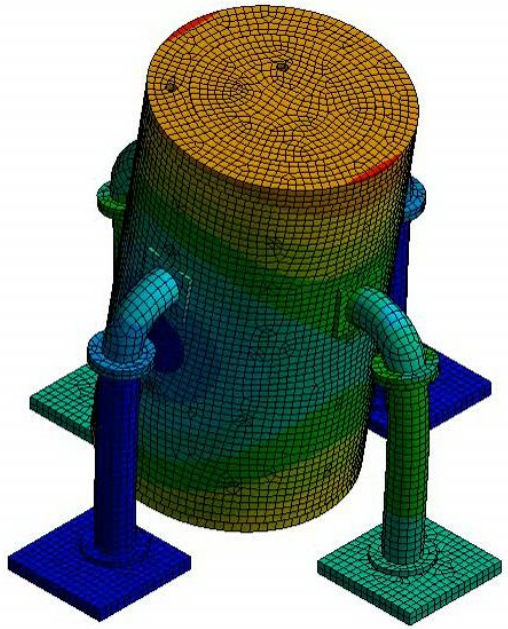
Finite Element,
Experimental
Modal Analysis of Reactor



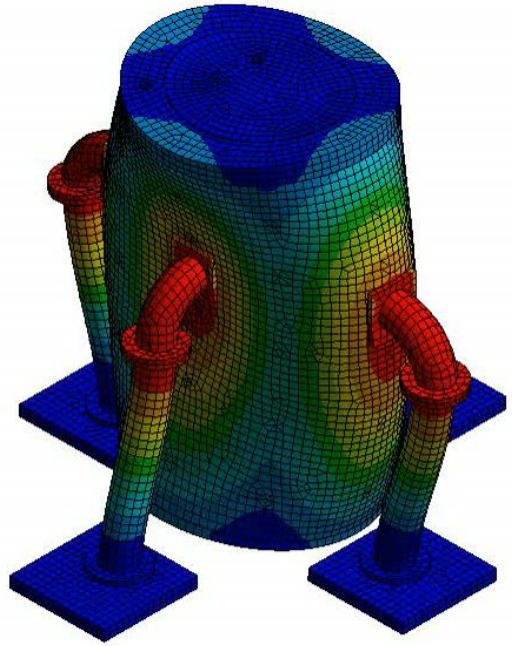
Model of Reactor



Vibration Motion of Reactor

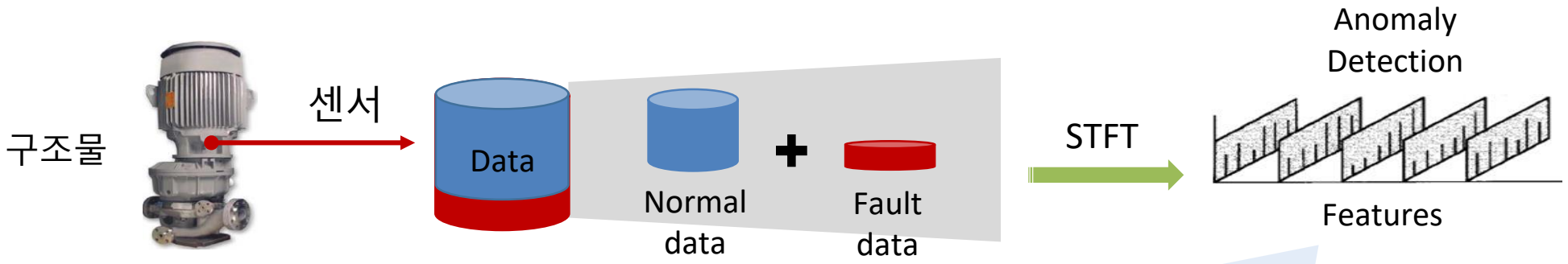


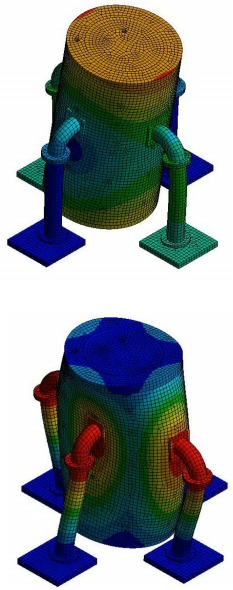
64.6 Hz

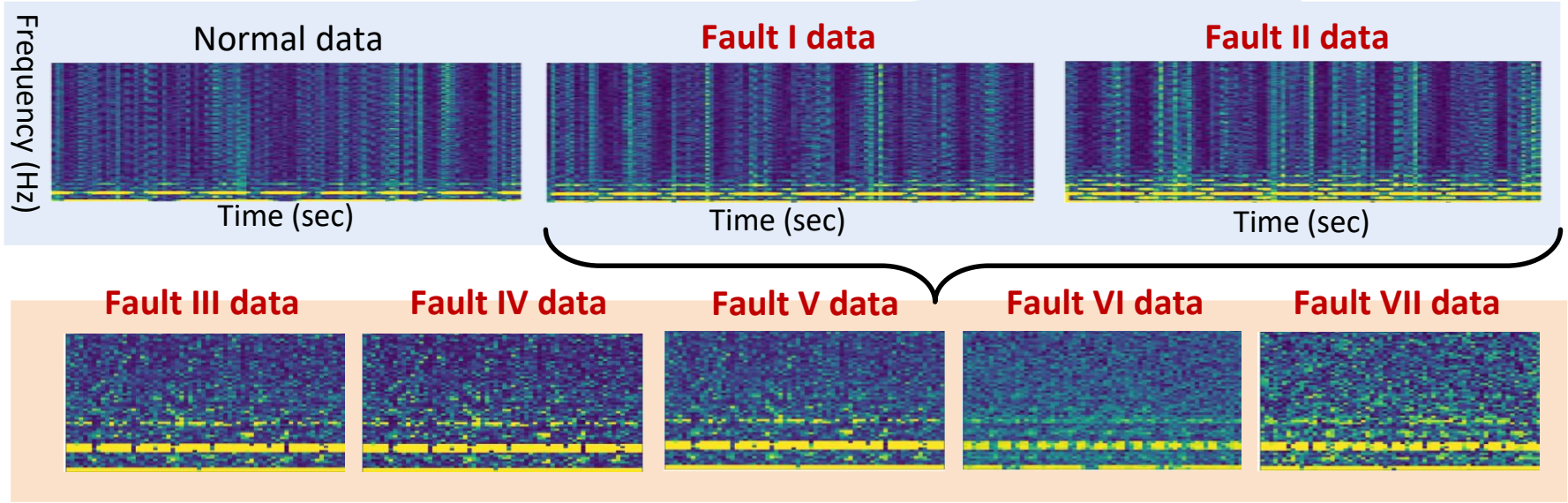


308.5 Hz

Hi-fidelity Physical Model based Anomaly Detection







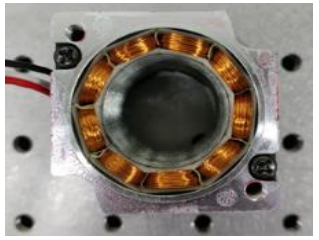
Real Data
측정

+

Fault Data
시뮬레이션

Deep learning based rotating machinery diagnosis using time-series data imaging

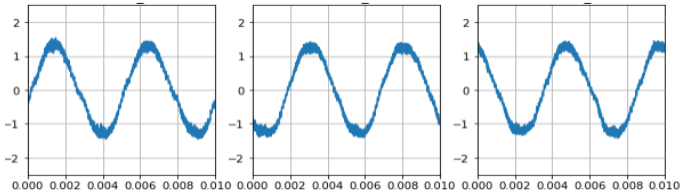
Data Acquisition

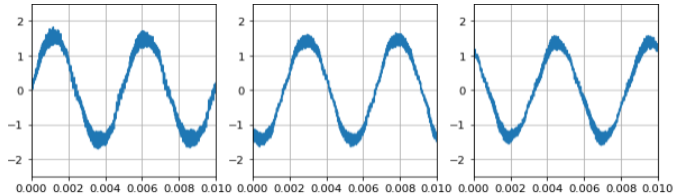


Normal



Abnormal

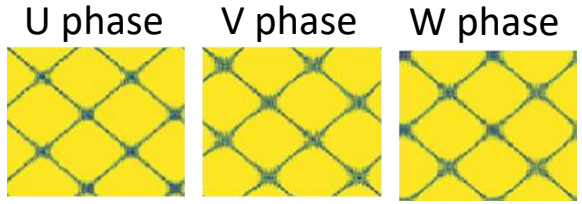




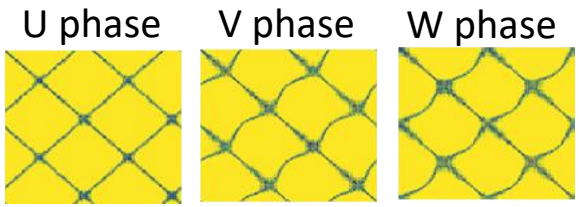
Time-Series Imaging



Normal

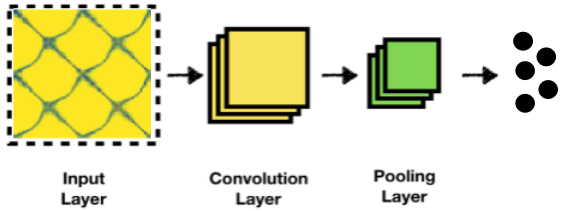


Abnormal 1

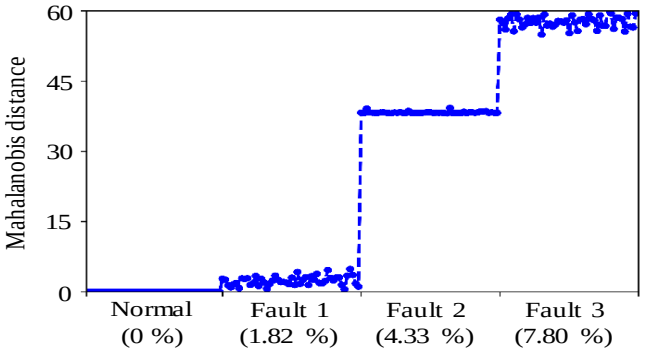


Abnormal 2

Unsupervised Learning



Feature extraction by CNN



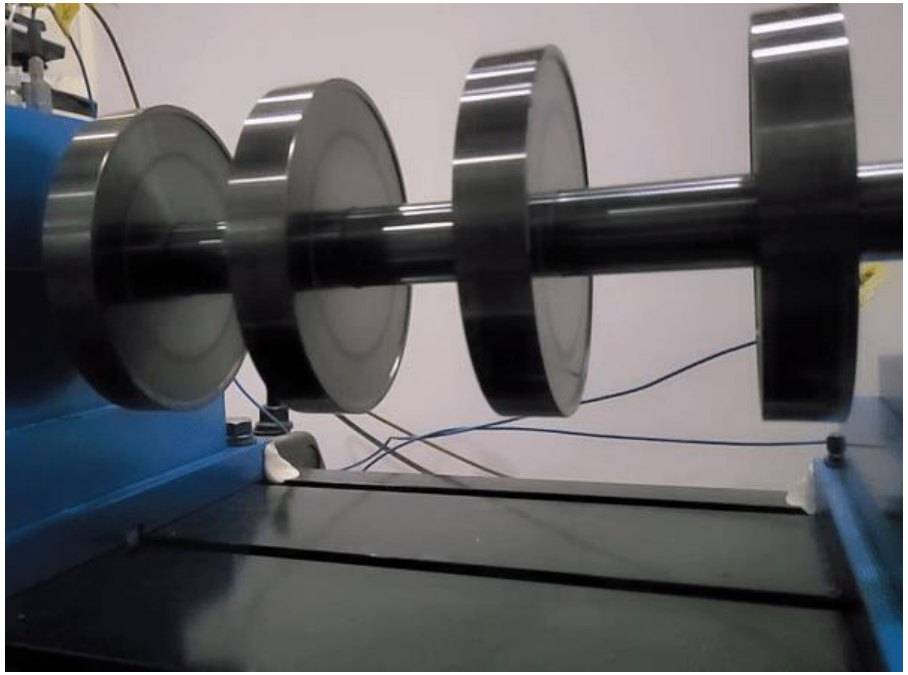
Status graph by MD*

*Mahalanobis distance

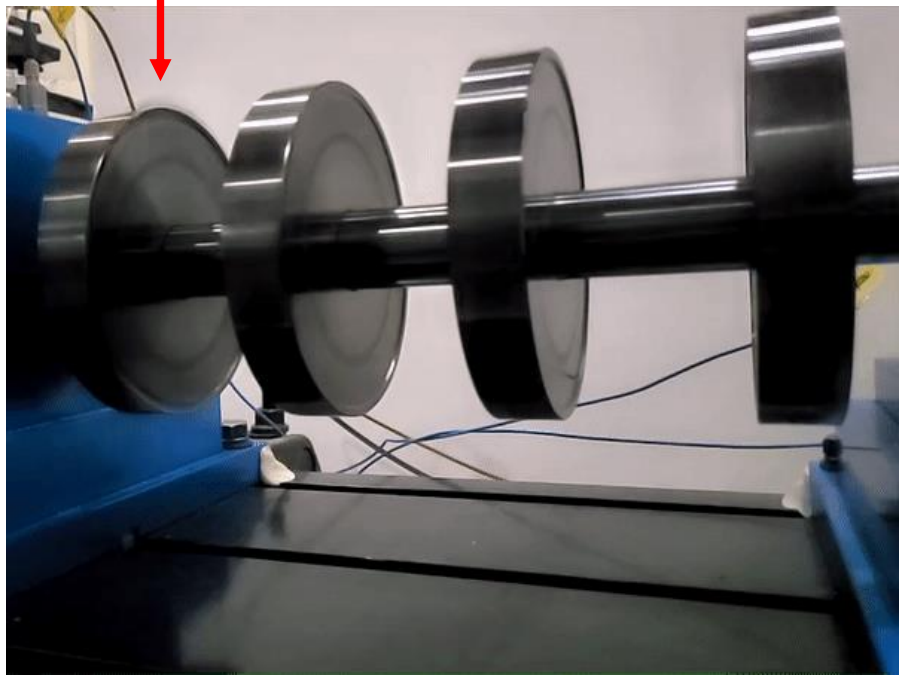
Best Paper Award, International Conference on Materials and Reliability, Korea, 2019.

Motion Amplification (Vibration Modal Analysis + Dynamic Vision Processing)

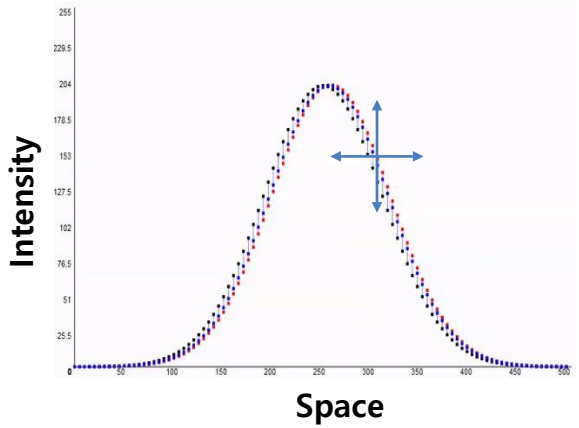
이상 부위



원본영상

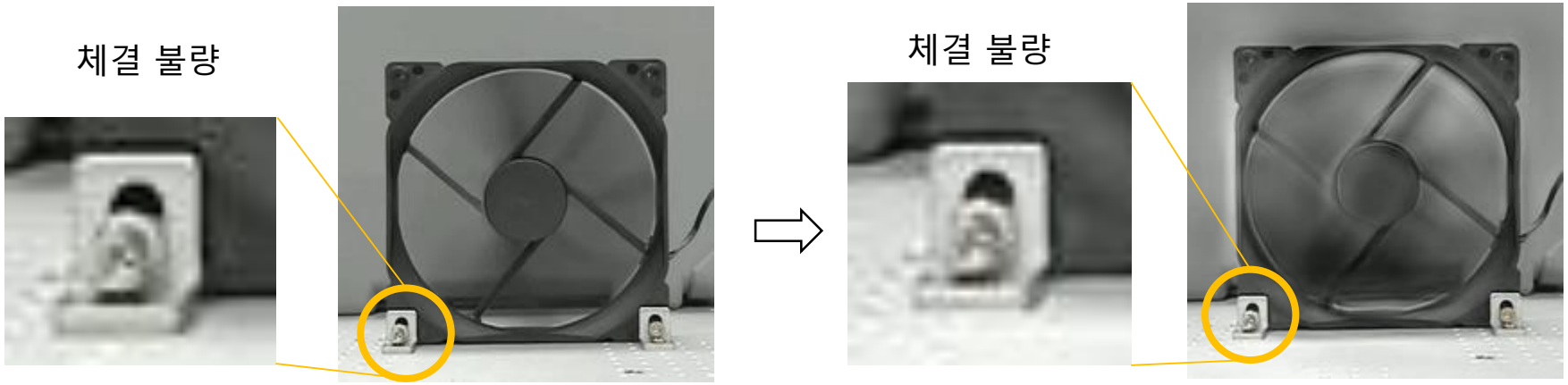


증폭영상



Motion Amplification (Vibration Modal Analysis + Dynamic Vision Processing)

10~20 Hz



원본영상 증폭영상

5~10 Hz



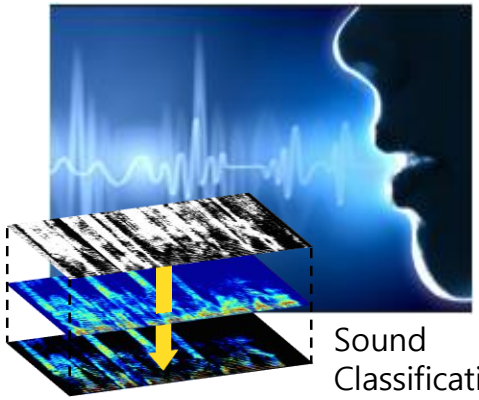
3D Shape and Motion Measurement (LIDAR)



3D 공간 Map

- High-definition, robust, compact 3D Lidar sensor system for autonomous vehicle and robots
- Time-of-Flight, Triangulation, Machine Learning based high-precision 3D information sensing
- A novel opto-mechanical signal processing methodology for depth extraction (AMCW, FMCW)
- 3D motion amplification using image signal processing

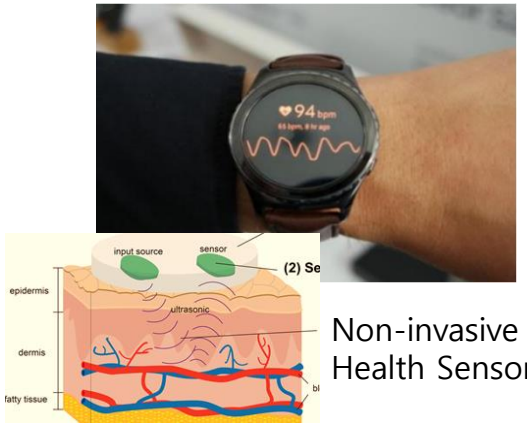
Acoustic Event/Source Recognition



Sound Classification/ Localization/ Separation

- Cocktail Party Problem: classification, localization and separation of the sound from real environment
- Sound Event Detection by Machine Learning algorithm
- Human auditory system (incl. HRTF) and vocal system analysis
- High-dynamic range, directional sound signal processing for robust and accurate sound recognition

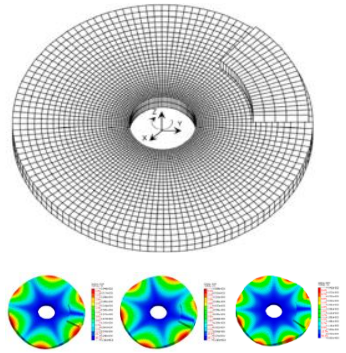
Biometric Recognition by Human Body Vibration



Non-invasive Health Sensor

- Non-invasive, wearable, continuous real-time health sensor for human body diagnosis (blood pressure, diabetes, skin etc.)
- Bio characteristics identification with human body vibration, coupled-field analysis and Machine Learning
- Virtual Patient System for CVDs
- A novel bio-signal processing scheme with Machine Learning for biomarker identification and health monitoring

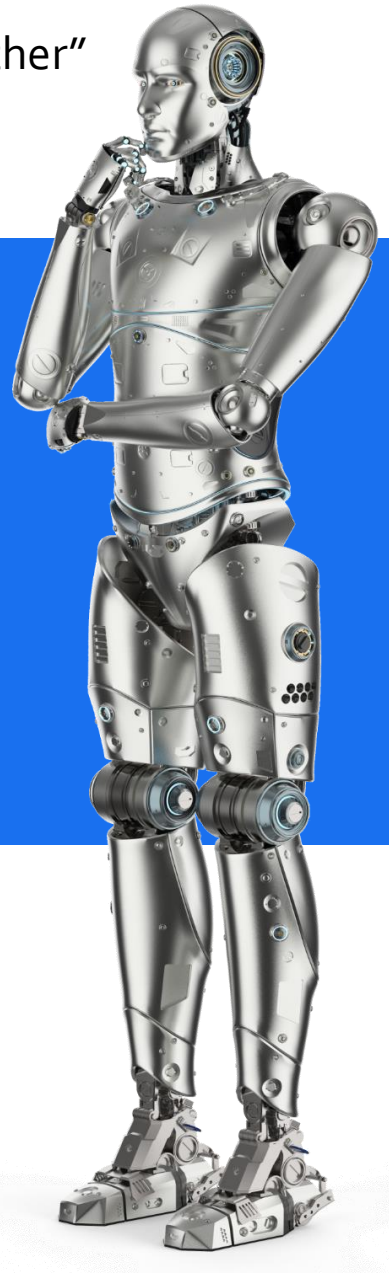
Vibration Modeling/Analysis, SHM(Structural Health Monitoring)



High Fidelity Structural Vibration Modeling and Analysis

- Coupled field dynamic analysis including fluid-structure, biosystems, sub-structured systems
- Uncertainty quantification
- Inverse Problems of structural vibration for identification
- Structural health monitoring
- Physical information and deep learning combined modeling
- High-fidelity dynamic Digital Twin

"Let's go further together"



KAIST HUMAN LAB

HUMAN-MACHINE INTERACTION

THANK YOU

Prof. Yong-Hwa Park, yhpark@kaist.ac.kr